# Evolving a multicellular worm from a DNA-encoded gene regulatory network

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## Summary

A population of simulated worms, each grown from one egg cell by a gene regulatory network read from a DNA string, evolved into a self-sustaining ecosystem of about 80 animals that feed, grow to 4 to 5 cells, expel waste and lay eggs. No behaviour was programmed; only the physics of the world and the reading rules for DNA were.

Three results stand out.

- **A growth-limiting signal evolved.** The worms use a diffusible gene product (S1) that represses cell division. Knocking it out lets bodies grow to 6.9 cells instead of 4.2 and cuts egg output by 79% (1.13 to 0.24 eggs per life).
- **The head marker became unnecessary.** A maternal marker in the egg was made to fade about 25 times faster over the course of evolution. The final worms reproduce equally well with the marker blocked (1.11 against 1.13 eggs per life).
- **Enemy avoidance did not evolve.** Predators cause about a third of deaths, yet blocking the worms' enemy senses changes nothing (1.15 against 1.13 eggs per life). Follow-up tests show that a close-range flee reflex would take over the population if it arose, so this is a failure of the search and not an absence of benefit.

This is a toy model. It shows what selection can build from a small rule set, and it says nothing quantitative about any real organism.

## 1. Introduction

The question was whether a body plan and a set of behaviours can be folded into one genome and unfolded again by development, in a model small enough to inspect gene by gene. Every cell of an animal carries the same DNA; cells differ because they express different genes, and gene expression is controlled by the products of other genes. That control system is the gene regulatory network (GRN).

The approach is inverse design by evolution. There is no formula that turns a target organism into a genome, so genomes are found by mutation and selection, and the result is then taken apart with knockouts, as a geneticist would.

The model draws on established ideas and makes no claim to novelty in its parts:

- GRNs as networks of continuous units with weighted activating and repressing inputs (Mjolsness, Sharp and Reinitz, 1991).
- Positional information from diffusing signals (Wolpert, 1969) and regulation by repressors and inducers (Jacob and Monod, 1961).
- Artificial embryogeny, in which a compact genome grows a multicellular body (Eggenberger, 1997; Bongard and Pfeifer, 2003; Stanley and Miikkulainen, 2003).
- Evolved virtual creatures scored on behaviour (Sims, 1994), and growth from a single cell under a shared local rule (Mordvintsev et al., 2020).

What this work adds is a single small system in which development, physiology, behaviour and reproduction are all controlled by one DNA-encoded network, together with a full record of where evolution stalled and why.

## 2. Model

A worm is a chain of up to 24 cells that starts as one egg cell. Every cell holds the same genome and its own expression state.

**Genome.** The genome is a string over A, C, G, T, about 900 letters at the start. A gene begins at the promoter `TATA`. The next 3 letters name its product, 2 letters set its default level, 1 letter sets how fast its level changes, and each following 6-letter block is a binding site: which regulator binds there and how strongly it activates or represses. A block starting with T ends the gene. Everything between genes is non-coding.

**Gene products.** A gene makes one of 22 products.

| Class | Count | Role |
|---|---|---|
| Transcription factors (T1 to T12) | 12 | Stay in the cell and regulate other genes |
| Signals (S1 to S3) | 3 | Spread to neighbouring cells and decay; S1 is long-range and slow, S3 short-range and fast |

| Class | Count | Role |
|---|---|---|
| Workers | 7 | mouth, digest, anus, left muscle, right muscle, divide, lay egg |

**Regulation.** Each tick, a gene's level moves toward a sigmoid of its default level plus the weighted sum of its regulators. Regulators are the 12 transcription factors, the 3 local signal concentrations and 9 senses: food smell, which side the food is on, enemy smell, which side the enemy is on, body energy, gut food, gut waste, the maternal marker, and whether the cell is at an end of the body.

**Cell and body.** A cell's five physiological workers share a fixed capacity, so no cell can do everything at full strength. Food enters at any cell expressing mouth, moves backward through the gut, is digested into energy and waste, and must be expelled or it blocks the gut and becomes toxic. Both muscles contracting push the worm forward; a left/right difference turns it. A cell divides when its divide product is high and the body has energy to spare; the daughter sits behind the mother and starts at half her levels.

**Symmetry breaking.** The egg carries a maternal marker that stays in the front cell and decays. The edge sense tells a cell it is at an end, but not which end.

**World.** A 500 by 500 wrap-around plane with food pellets in patches that grow back, and six predators that chase worms in proportion to their size. A bite on the head kills; a bite elsewhere removes that cell and everything behind it.

**Reproduction and mutation.** A worm of at least 5 cells with enough energy lays an egg when its lay product is on. The egg carries the parent's DNA with copying errors: letter substitutions (0.2% per letter), small insertions and deletions, and occasional duplication or loss of a 30 to 200 letter stretch. An egg whose DNA lacks a gene for any of the seven workers does not hatch.

## 3. Evolution procedure

Evolution ran in two stages, because a shared world started from random DNA never produced a worm that could reproduce.

**Stage 1, scored warm-up.** 300 random genomes, each with one gene per product and random regulation. Each genome was grown alone from an egg in 6 random worlds for 3,000 ticks and scored as energy gained plus 100 per egg laid. The top 15 were kept; the rest were replaced by mutated copies of tournament winners. This ran for 450

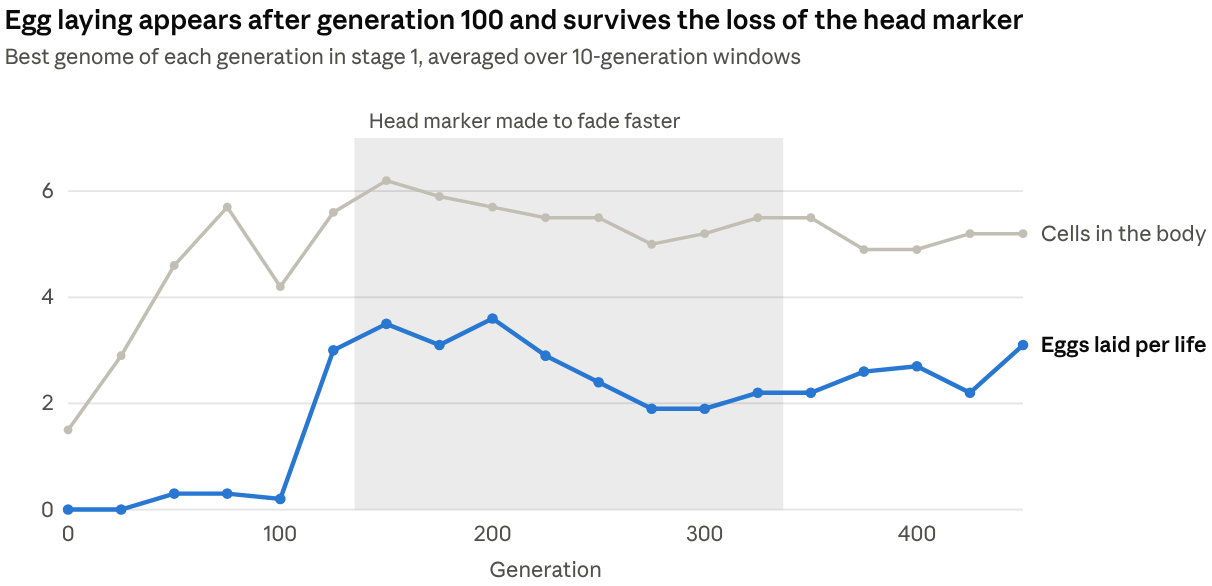
generations with a mutation rate three times that of stage 2. The maternal marker's half-life was held at 3,500 ticks for the first 135 generations, then shortened geometrically to 138 ticks by generation 337.

**Stage 2, no score.** The 40 best stage 1 genomes were released, two copies each, into four independent shared worlds for 200,000 ticks. Nothing is scored. A lineage spreads only by laying eggs that hatch and survive. Hatching is less likely in a crowded world. If a world drops below 20 worms, eggs are added from outside; this never happened in the reported run.

**Testing.** The final population was tested as a geneticist tests a mutant. For each condition a fresh world was stocked with the final genomes, mutation was switched off, one sense was blocked or one gene product was forced to zero in every worm, and the world ran for 20,000 ticks. The main measure is eggs laid per completed life.

# 4. Results

## 4.1 Stage 1: bodies first, then eggs




evolved.json, stage 1 history · 450 generations, 300 genomes each, 6 worlds per genome

Multicellular bodies appeared within 30 generations. Egg laying became established between generations 100 and 125. Shortening the marker's half-life from 3,500 to 138 ticks cost about half the egg output at its worst, near generation 300, and the population then recovered with the marker at its final setting.

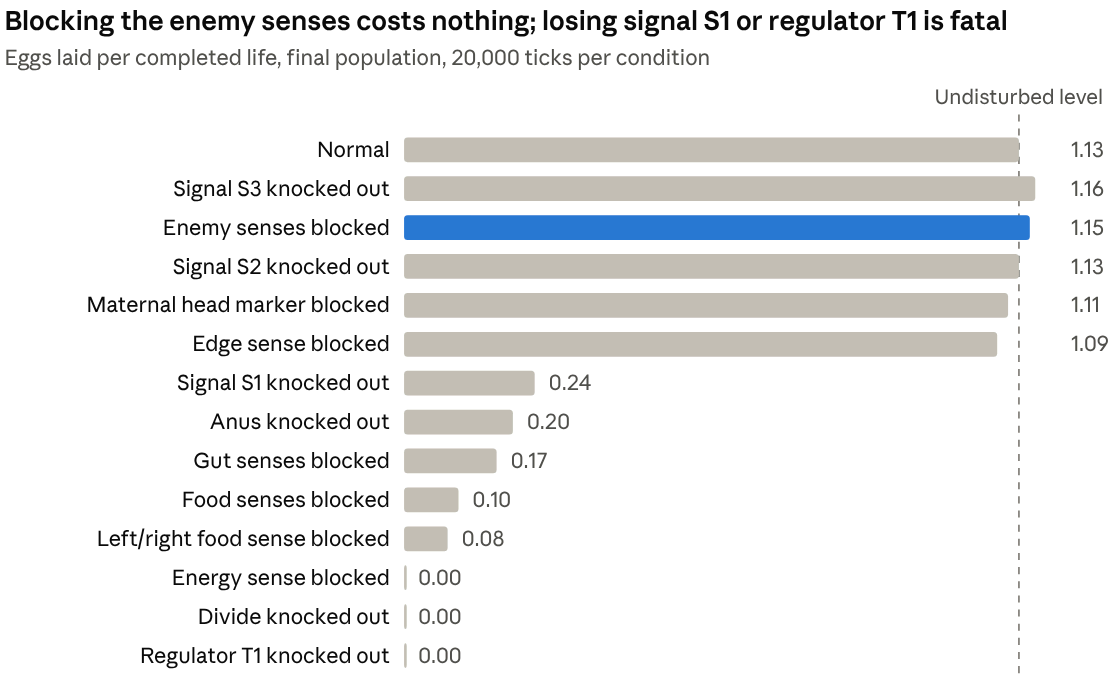
## 4.2 Stage 2: a self-sustaining population

All figures are ranges over ten samples taken every 20,000 ticks in the reported world.

| Measure | Value |
|---|---|
| Worms alive | 77 to 95 |
| Cells per worm | 4.2 to 4.6 |
| Eggs hatched per 5,000 ticks | 102 to 129 |
| Eggs added from outside | 0 |
| Eggs laid per life | 1.2 to 1.4 |
| Mean lifespan | 3,400 to 4,100 ticks (the age limit is 6,000) |

Of the deaths in those samples, 37% were from old age, 34% from starvation and 29% from predators. The longest lineage at the end was 76 generations deep.

## 4.3 What the worms depend on




analyze.js output · final population at tick 200,000, mutation off

Without S1, worms grew to 6.9 cells instead of 4.2 and 70% starved. Without the energy sense they stayed at 1.8 cells and never laid. Eleven of the twelve transcription factor kinds could be removed one at a time with no measurable effect (1.11 to 1.21 eggs per life); T1 could not. With the enemy senses blocked, predators caused 34% of deaths against 32% normally.

## 4.4 The evolved circuits

The wiring below is read from one representative genome of the final population, which has 10 genes. Strengths are in model units, where 4.5 is the maximum.

| Function | Wiring found | Reading |
|---|---|---|
| Body size | divide is on by default, activated by energy at two binding sites, repressed by S1 (4.5). S1 is on by default. | Cells divide while energy is high, until slowly accumulating S1 shuts division off |
| Waste | T1 is on by default and repressed by gut waste (4.5). anus is repressed by T1 (4.5). | A double negative: waste switches T1 off, which releases the anus gene |
| Steering | Right muscle is activated by food on the right (4.5); left muscle is repressed by it (4.5) | The worm turns toward food |
| Feeding | mouth is on by default, activated by the edge sense, repressed by gut waste, energy and S1 | Eats at the body's ends and stops when full |
| Reproduction | lay is on by default and repressed by the edge sense | Interior cells trigger laying, subject to the 5-cell and energy rules |

Two of these have the same logic as textbook biology, by analogy only. Growth limited by an inhibitor that the tissue itself makes is how myostatin limits muscle. A gene held off by a repressor that an input removes is how the lac operon and the bacterial SOS response work.

## 4.5 Genome size

Random starting genomes had 22 genes in about 920 letters. The stage 1 founders averaged 11.7 genes in 1,581 letters, and the final population 10.4 genes in 1,611 letters. Unused regulator genes were lost while non-coding DNA grew. Every final genome kept T1 and S1. 65% also carried an S3 gene, although removing S3 had no measurable effect.

## 4.6 Earlier experiment with a fixed gene set

A first version used a fixed set of 19 genes stored as a weight matrix, permanent head and tail markers, one worm per world, and energy gained as the score. After 300 generations worms reached 9 cells and gained 298 energy per life, against 11 for a random genome. Blocking the left/right food sense cut that to 20 and removing the pole markers cut it to 0. Enemy senses had no effect there either (299), and no signal gene was used. Those two gaps motivated the changes in the model described here.

## 4.7 Why enemy avoidance did not evolve

A close-range flee reflex is strongly favoured in this world, a long-range one is strongly disfavoured, and evolution found neither. Four follow-up experiments on the final population established this.

**A reflex added by hand.** Every worm was given a hard-wired rule: turn the head away whenever enemy smell exceeds a threshold. With the threshold at 0.15, which is reached about 58 units from a predator, deaths from predators fell from 33% to 0%. Eggs per life did not change (1.16 in both cases), because a full world holds egg output per life near replacement whatever the worms do. Egg output per life is therefore a poor measure of advantage, and the next tests use competition instead.

**Head to head.** Half the starting worms and all their descendants carried the reflex; both kinds shared one world with mutation off. Each cell below is the reflex carriers' share of the population after 80,000 ticks in four worlds, starting from 50%.

| Reflex triggers within | 6 predators | 24 predators |
|---|---|---|
| About 58 units (threshold 0.15) | 14%, 6%, 31%, 20% | 0%, 0%, 0%, 0% |
| About 35 units (threshold 0.5) | 100%, 100%, 91%, 100% | 100%, 100%, 100%, 100% |
| About 20 units (threshold 0.8) | 94%, 78%, 78%, 99% | 100%, 100%, 100%, 100% |

Fleeing at long range loses, most likely because the worm spends its time turning away from distant predators instead of feeding. Fleeing only at close range wins in all 16 worlds.

**More predators, more time.** The final population was left to evolve for a further 400,000 ticks with 24 predators instead of 6. It did not adapt. Predators caused 83% to 87% of deaths throughout, the population fell to the refill floor of about 22 worms, and roughly 300 eggs per 50,000 ticks had to be added from outside. Tested afterwards with mutation off, these worms did no better than their ancestors (82% of deaths from predators in both), and blocking their enemy senses made no clear difference (84% against 82%).

**Interpretation.** This part is a hypothesis. The useful behaviour is conditional: turn away only when the enemy is close, and let that override the turn toward food. In this model a gene sums its inputs, so the simplest wiring, an enemy-side sense acting directly on a muscle gene, gives the always-on reflex that the competition shows to be harmful. The beneficial version needs a second input to gate the first, and there may be no sequence of single mutations that improves at every step. Every genome in the high-predation population did carry at least one enemy-sense binding site, and one inspected genome had the enemy-side sense activating the correct muscle, but without measurable effect on survival.

## 5. Where evolution stalled, and the rules added

The reported run was the eighth attempt. Each earlier one stalled for a specific reason, and each fix is a design decision by the authors, not something that evolved.

| Stall | Cause found | Rule added |
|---|---|---|
| No worm ever reproduced in a shared world started from random DNA | Random worms starve or are eaten before any lays an egg, so there is nothing to select | A scored warm-up stage before the shared world |
| Worms stayed single-celled and laid one egg per life | Hoarding energy for an egg scored higher than building a body | Only worms of 5 or more cells can lay |
| No bodies after 120 generations, even with a long-lived marker | The unused divide gene was deleted within 30 generations and could not be regained | An egg hatches only if its DNA has a gene for each of the seven workers |

| Stall | Cause found | Rule added |
|---|---|---|
| Bodies stuck at 2 cells for 150 generations | The energy budget: larger bodies ate the same and cost more (table below) | Pellets 33% richer, muscle cost down 42%, resting cost down 25% |
| Progress about 25 times slower than with the fixed gene set | A point mutation matters only if it hits the few letters that set a level or a binding site | Mutation rate tripled in stage 1 |
| Shared world not self-sustaining | The world was refilled up to 100 worms, more than the food could feed | Refill floor lowered to 20, food regrowth doubled |

One run also hung on a software fault, a food pellet filed under one grid square and looked up under its neighbour. That was a bug, not a finding.

The energy-budget stall was measured directly by forcing a generation-190 worm to grow to fixed sizes, 40 trials each, under the earlier costs.

| Forced body size | Energy gained per life | Mean lifespan (ticks) | Starved, of 40 |
|---|---|---|---|
| 1 cell | 53 | 2,303 | 0 |
| 2 cells (the evolved size) | 102 | 2,225 | 4 |
| 3 cells | 107 | 1,897 | 18 |
| 4 cells | 86 | 1,297 | 25 |
| 6 cells | 65 | 861 | 35 |

Evolution had stopped at the size the world could support. The general lesson is that each stall looked like a failure of the search and turned out to be a correct response to the rules as written.

## 6. Limitations

- **One lineage.** All reported results come from one stage 1 run with one random seed and one of four stage 2 worlds. Whether the S1 growth limiter would evolve again is untested.

- **Hand-set rules shaped the outcome.** The 5-cell rule for laying, the essential worker genes and the energy budget were chosen to get past stalls. A different choice could give a different animal.
- **The body is a one-dimensional chain.** There is no shape, no mechanics and no cell movement, so nothing here bears on morphogenesis.
- **The network is shallow.** Final genomes have about 10 genes and one essential transcription factor. Most senses act directly on worker genes.
- **Behaviour is reflexive.** There is no nervous system and no learning within a life.
- **Reproduction is asexual.** There is no recombination in stage 2.
- **The edge sense is a given.** Worms no longer need the maternal marker, but they were always able to sense the ends of the body. Blocking the edge sense alone had little effect (1.09 eggs per life); blocking both together was not tested.
- **No statistics.** Each test condition was run once for 20,000 ticks. Differences of a few percent between conditions are within noise; the collapses to near zero are not.

## 7. Next steps

1. Get close-range enemy avoidance to evolve unaided, for example by letting one binding site require two regulators together, or by running far longer.
2. Repeat stage 1 with several seeds to see which circuits recur.
3. Block the maternal marker and the edge sense together, to test whether head identity is truly self-maintained.
4. Give the body a second dimension so that shape can evolve.
5. Add sexual recombination to the shared world.

## 8. Reproducibility

The simulation is under 1,000 lines of dependency-free JavaScript and runs locally under Node.js; the reported run took about 16 minutes on a 10-core laptop. The code was written by Claude Fable 5.2 (Anthropic) working in Claude Code, which also ran the experiments and drafted this report. Hamid Mohamadlou posed the questions and directed the work. The follow-up experiments are in exp_enemy.js, exp_enemy2.js and exp_enemy3.js.

| File | Purpose |
|---|---|
| `sim.js` | World, worm, DNA parser, mutation |

| File | Purpose |
|---|---|
| `evolve.js` | Stage 1 and stage 2 |
| `analyze.js` | Sense-blocking and knockout tests |
| `index.html` | Live viewer: world, per-cell gene activity, gene list, DNA |

```
node evolve.js 450 200000 1
node analyze.js 20000
```

Runs are deterministic for a given seed and parameter set.

## References

Cited from memory; check each before formal publication.


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